

Gradient Method (Sample for the project report)

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Abstract

In this project, we study the gradient descent method for solving unconstrained smooth optimization problems. We begin by presenting its formulation and discuss its broad range of applications across fields such as machine learning, signal processing, and statistics. The gradient method is introduced as an iterative algorithm that follows the direction of steepest descent to minimize a differentiable objective function. We describe the method in detail, including its convergence properties under Lipschitz continuity and the commonly used stopping criterion based on the gradient norm. To evaluate its practical performance, we apply the method to a least-squares problem and compare the results using different step size strategies, including fixed step size, diminishing step size, and backtracking line search.

Keywords: Convex Optimization, Gradient Method, Least-Square Problem

1. Introduction

Optimization plays a fundamental role in many fields of engineering, data science, and applied mathematics. At its core, optimization is concerned with finding the best solution—according to some criterion—from a set of possible choices. One of the most widely studied and applied optimization problems is the **unconstrained minimization** of a smooth function. In this context, the **gradient method** (also known as gradient descent) offers a simple yet powerful iterative technique to reach a local minimum of a continuously differentiable function.

In this work, we solve the following minimization problem using only the first-order information:

$$\min_x f(x),$$

where $f(x): \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex, i.e., $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$, and continuously differentiable function with Lipschitz continuous gradient, i.e., there exists $L > 0$ such that $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$.

A classic example of such a function is the **least-squares** objective used in linear regression, where the goal is to fit a model to data in a way that minimizes the sum of squared errors. Although solving the least-squares problem, $\min \|Ax - b\|^2$, directly via matrix inversion is possible, such an approach can be computationally expensive or unstable in high-dimensional settings. The gradient method circumvents these issues by iteratively updating the solution based on the gradient information, requiring only matrix-vector multiplications. This makes it especially appealing for large-scale problems where efficiency is critical. This project introduces the gradient method, establishes its theoretical foundations, analyzes its convergence behavior, and provides practical examples to illustrate its performance under different step-size strategies.

1.1 Applications

The gradient method, particularly in its basic form, finds widespread applications across a broad range of fields:

- **Machine Learning and Data Science:**
In supervised learning, especially linear and logistic regression, the objective is typically a loss function over data, which is minimized using gradient-based techniques. Gradient descent and its variants form the backbone of most modern machine learning algorithms.
- **Signal Processing:**
Gradient methods are used for denoising, filtering, and reconstruction of signals, where the optimization problem often involves minimizing an energy function that quantifies distortion.
- **Statistics and Estimation:**
Maximum likelihood estimation and other statistical fitting techniques often reduce to optimization problems solvable via gradient descent.
- **Economics and Finance:**
Optimization models in portfolio selection, risk minimization, and game-theoretic settings commonly rely on gradient-based updates for finding equilibrium or optimal strategies.
- **Engineering Design and Operations Research:**
Engineering problems such as structural optimization, resource allocation, and network design often require minimizing objective functions that are smooth and amenable to gradient methods.

These applications highlight the versatility and practical relevance of gradient methods, especially in scenarios where closed-form solutions are impractical or nonexistent.

1.2 Literature review

Gradient descent is a foundational first-order method used for solving unconstrained optimization problems where the objective function is differentiable. The algorithm operates by iteratively updating the decision variable in the direction opposite to the gradient, exploiting the fact that this is the direction of steepest decrease. While the concept of following the gradient dates back to the 19th century—initially proposed by Augustin-Louis Cauchy in 1847 [1] and later echoed by Jacques Hadamard [2] in the early 20th century—the formal analysis of its convergence in nonlinear settings was pioneered by Haskell Curry in 1944 [3]. Since then, gradient descent has been extensively studied and generalized to various forms, including its modern applications in large-scale machine learning, where it is commonly used to minimize loss functions.

In recent decades, several key texts have expanded upon the theoretical and algorithmic properties of gradient methods. Beck provides a comprehensive introduction to nonlinear optimization, presenting convergence theory, implementation strategies, and MATLAB and Python-based examples [4]. Nocedal and Wright offer a broad and detailed treatment of numerical optimization, emphasizing descent algorithms and practical performance considerations [5]. Boyd and Vandenberghe contextualize gradient descent within the framework of convex analysis and duality theory [6]. For a more complexity-oriented perspective, Bubeck [7] surveys convergence rates and computational lower bounds for first-order methods, including gradient descent, providing insight into their theoretical limitations. Lastly, Bottou, Curtis, and Nocedal [8] focus on the application of gradient methods in large-scale machine learning, discussing challenges like noise, data sparsity, and the need for adaptive step-size techniques. Together, these works form a robust theoretical and practical foundation for understanding and applying the gradient method in modern optimization tasks.

2. Methodology/Algorithm description

We consider minimizing a smooth, unconstrained convex function $f(x)$. The gradient method iteratively updates the decision variable as:

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k)$$

where $\alpha_k > 0$ is the step size and $\nabla f(x_k)$ is the gradient at iteration k . If $\nabla f(x_k) \neq 0$, then $-\nabla f(x_k)$ is the steepest descent direction.

Convergence Results:

- If the gradient of f is Lipschitz continuous with constant L , then using a constant step size $\alpha < 2/L$ guarantees that $\|\nabla f(x_k)\| \rightarrow 0$ as $k \rightarrow \infty$.
- If f is convex and $\alpha = 1/L$, the function values converge at the rate $O(1/T)$, where T is the number of iterations, i.e., $f(x_T) - f(x^*) \leq \frac{L}{2T} \|x_0 - x^*\|^2$, where x^* is the optimal solution, x_0 is the initial point and L is the Lipschitz constant of the gradient.

Stopping Criteria:

If f is **convex** and differentiable, then **every local minimum is also a global minimum**. This property is particularly powerful when using first-order methods such as the **gradient method**. Specifically, if the algorithm reaches a point x^* such that $\nabla f(x_k) = 0$, then x^* is guaranteed to be a **global minimizer** of the function f . This guarantees that, under convexity, as long as the gradient method converges (i.e., $\|\nabla f(x_k)\| \rightarrow 0$), it will yield a globally optimal solution. Therefore, a natural and widely used **stopping criterion** in this setting is:

$$\|\nabla f(x_k)\| \leq \epsilon,$$

where $\epsilon > 0$ is a small tolerance. Once this condition is met, we can confidently terminate the algorithm, knowing that the current iterate x_k is sufficiently close to a global minimum of the objective function.

3. Numerical Experiments

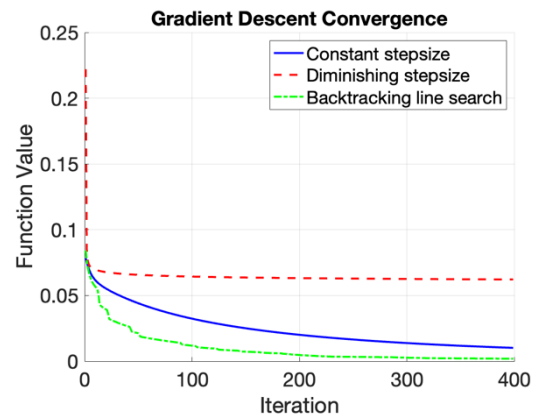
In this section, we solve the least squares optimization problem

$$\min_x f(x) := \|Ax - b\|^2$$

using gradient descent with different step-size strategies. A is an $m \times n$ matrix with entries drawn uniformly at random, x is an $n \times 1$ decision variable, b is an $m \times 1$ vector with entries drawn uniformly at random. We set $m = n = 5$, $x_0 = \mathbf{0}$ and we terminate the method after 400 iterations.

The gradient of the objective function is $\nabla f(x) = 2A^T(Ax - b)$ and Lipschitz constant of the gradient is $L = 2\|A\|^2$. we compare the performance of the gradient method under three step size strategies:

a fixed step size $\alpha = 1/L$, a diminishing stepsize $\alpha = 0.1/k$, and backtracking line search. Among these, the backtracking line search demonstrates the fastest convergence, highlighting its effectiveness in adapting the step size dynamically to ensure rapid progress toward the minimum.



4. Conclusion and Future Direction

We presented the gradient method as a simple yet effective algorithm for unconstrained smooth optimization. Its effectiveness depends on appropriate step size selection and the smoothness of the objective function. Future work may involve acceleration methods such as momentum or adaptive learning rates, and extensions to constrained or stochastic optimization problems.

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